

Existence of a Model Companion for Groups of Exponent 3

Yawara Ishida, Ryosuke Mizuno and Kota Takeuchi

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Abstract

In this article, we prove that the theory T_3 of groups of exponent 3 has a model companion. Previous work established the existence of model companions for theories of groups of fixed finite exponent and nilpotency class at most 2 (Saracino-Wood), and for theories of groups of prime exponent p and nilpotency class at most $c < p$ (Maier). These results do not cover T_3 , since groups of exponent 3 may have nilpotency class 3. Our theorem verifies the $n = 3$ case of Takeuchi's conjecture that, for each integer $n > 1$, the theory T_n of groups of exponent n has a model companion if and only if every finitely generated group of exponent n is finite, or equivalently, if and only if the Burnside problem has a positive solution for exponent n .

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1 Introduction

The history of research on model companions of groups. Takeuchi's conjecture and its motivation. Main results and further work.

2 Preliminaries

Throughout, groups are regarded as structures in the language $\mathcal{L}_{\text{grp}} = \{\cdot, {}^{-1}, 1\}$. Let T_{grp} denote the theory of groups. For an integer $n > 1$, set

$$T_n := T_{\text{grp}} \cup \{\forall x (x^n = 1)\}.$$

Thus, by a group of exponent n , we mean a group whose exponent divides n .

We fix the following notation for group elements:

Notation 2.1. 1. $[a, b] = aba^{-1}b^{-1}$

2. $[a_1, \dots, a_k] = [[a_1, \dots, a_{k-1}], a_k]$
3. $a^b = b^{-1}ab$
4. $\mathcal{V}_3$ is the variety of groups satisfying $x^3 = 1$; equivalently, $\mathcal{V}_3 = \{M \mid M \models T_3\}$.
5. $G * H$ is the free product of groups G and H .
6. For a subset S of a group G , $\langle\langle S \rangle\rangle_G$ denotes the normal closure of S in G . We omit the subscript when the ambient group is clear.
7. $d(G)$ is the minimum number of generators of G .
8. $G *_{\mathcal{V}_3} H$ is the coproduct of G and H in the variety $\mathcal{V}_3$. Equivalently,

$$G *_{\mathcal{V}_3} H = (G * H) / \langle\langle (G * H)^3 \rangle\rangle_{G * H}.$$

9. For a set X , $F_{\mathcal{V}_3}(X)$ is the free group in the variety $\mathcal{V}_3$ generated by X . Equivalently, if $F(X)$ is the free group on X , then

$$F_{\mathcal{V}_3}(X) = F(X) / \langle\langle F(X)^3 \rangle\rangle_{F(X)}.$$

If $|X| = r \in \mathbb{N}$, then $F_{\mathcal{V}_3}(X)$ is called the Burnside group $B(r, 3)$.

2.1 Model-theoretic background

Let T and T' be L -theories.

Definition 2.2. 1. T_0 and T_1 are called companions if for $i = 0, 1$, for every $M_i \models T_i$ there is $M_{1-i} \models T_{1-i}$ such that $M_i \subset M_{1-i}$.

2. T is said to be model complete if for every formula $\theta(\bar{x})$ there is an existential formula $\psi(\bar{x})$ such that $T \models \forall \bar{x}(\theta(\bar{x}) \leftrightarrow \psi(\bar{x}))$.
3. T^* is called a model companion of T if T^* is a companion of T and is model complete.
4. $M \models T$ is an existentially closed model (e.c. model) of T if for any $\bar{a} \in M$ and any extension $M \subset N \models T$, $N \models \exists \bar{x}\varphi(\bar{x}, \bar{a})$ implies $M \models \exists \bar{x}\varphi(\bar{x}, \bar{a})$ where $\varphi(\bar{x}, \bar{y})$ is quantifier free.
5. T is a Π_2 -theory if there is a set T' of $\forall\exists$ -sentences such that $T \equiv T'$.
6. Let $M \models T$ and let $A \subset M$ be a finite subset. Set

$$\text{Diag}_A(\bar{x}) = \{\theta(\bar{x}) \mid M \models \theta(A), \theta(\bar{x}) \text{ is quantifier-free}\},$$

where $|\bar{x}| = |A|$. If L is finite and T is locally finite, then $\text{Diag}_A(\bar{x})$ is finite modulo T .

Fact 2.3. Let T be a Π_2 -theory. Then for an L -theory T^* , T^* is a model companion of T if and only if

$$\{M \mid M \models T^*\} = \{M \models T \mid M \text{ is an e.c. model of } T\}.$$

Definition 2.4. Let T be an L -theory. We say T is locally finite if, for every $M \models T$ and every finite subset $A \subset M$, the substructure $\langle A \rangle \subset M$ generated by A is finite.

The next fact follows from a standard compactness argument.

Fact 2.5. Let L be finite and let T be a locally finite L -theory. Then for any $n \in \mathbb{N}$, there is $m \in \mathbb{N}$ such that if $A \subset M \models T$ is a subset with $|A| \leq n$, then $|\langle A \rangle| \leq m$.

Fact 2.6. Let L be finite and let T be a locally finite L -theory. Suppose that T is a Π_2 -theory. Then the following are equivalent.

1. T has a model companion T^* .
2. For any finite substructures $A \subset B$ of a model $M_0 \models T$, there is $n_{A,B} \in \mathbb{N}$ satisfying the following: Let M be an e.c. model of T and let $A \subset M$. If M and B don't have an amalgam over A in T , then there is $A \subset C \subset M$ such that
 - (a) C is generated by at most $n_{A,B}$ elements as an L -substructure,
 - (b) C and B don't have an amalgam over A in T .

Proof. For given finite $A \subset B$, let $\theta_{B,A}(\bar{x}, A)$ be a quantifier-free $L(A)$ -formula expressing $\text{Diag}_{AB}(\bar{x}, \bar{y})$; in other words, $\theta_{B,A}(B', A')$ holds if and only if $A \cong A'$ and $B \cong_A B'$ after identifying A' with A . Since L is finite, we can find such a formula $\theta_{B,A}(\bar{x}, A)$.

(1 $\Rightarrow$ 2) Since T^* is a model companion, $T^* \models \forall \bar{y}(\neg \exists \bar{x} \theta_{B,A}(\bar{x}, \bar{y}) \leftrightarrow \exists \bar{z} \psi(\bar{z}, \bar{y}))$ for some quantifier-free L -formula $\psi(\bar{z}, \bar{y})$. Let $n_{A,B} = |\bar{z}| + |A|$.

Suppose M is an e.c. model of T and $A \subset M$. If M and B don't have an amalgam over A in T , then $M \models \neg \exists \bar{x} \theta_{B,A}(\bar{x}, A)$. Since $M \models T^*$, $M \models \exists \bar{z} \psi(\bar{z}, A)$. Let $\bar{c} \in M$ be a realization of $\psi(\bar{z}, A)$ in M and let $C = \langle \bar{c}, A \rangle_M \subset M$. Suppose that C and B have an amalgam $N_0 \models T$ over A . We can extend $N_0 \subset N \models T^*$. Then $N \models \psi(\bar{c}, A)$, hence $N \models \neg \exists \bar{x} \theta_{B,A}(\bar{x}, A)$. However, N contains a copy of B over A , which implies $\exists \bar{x} \theta_{B,A}(\bar{x}, A)$.

(2 $\Rightarrow$ 1) For each finite $A \subset B$, let $K(A, B)$ be the set of all (isomorphism types of) $C \supset A$ over A such that

1. C is a substructure of some $N \models T$ and C is generated by at most $n_{A,B}$ elements,
2. C and B don't have an amalgam in T over A .

Because T is locally finite, the size $|C|$ is uniformly bounded by some $m_{A,B} \in \mathbb{N}$. Hence $K(A, B)$ is finite since L is finite. Let $\theta_A(\bar{y})$ be a quantifier free formula expressing $\text{Diag}_A(\bar{y})$, and let

$$T^* = T \cup \left\{ \forall \bar{y} \left[\theta_A(\bar{y}) \wedge \left(\bigwedge_{C \in K(A,B)} \neg \exists \bar{z} \theta_{C,A}(\bar{z}, \bar{y}) \right) \rightarrow \exists \bar{x} \theta_{B,A}(\bar{x}, \bar{y}) \right] \mid A \subset B \text{ are finite} \right\}.$$

If $M \models T$ is an e.c. model, then clearly $M \models T^*$, because $A \subset M$ and $A \subset B$ have an amalgam in T over A if and only if $M \models \exists \bar{x} \theta_{B,A}(\bar{x}, A)$. Conversely, suppose $M \models T^*$ and let $\bar{a} \subset M \subset N \models T$, $\varphi(\bar{x}, \bar{a})$ a quantifier-free formula, and $N \models \exists \bar{x} \varphi(\bar{x}, \bar{a})$. Then there is $\bar{b} \in N$ such that $N \models \varphi(\bar{b}, \bar{a})$, so let $B = \langle \bar{b}, \bar{a} \rangle_N$ and $A = \langle \bar{a} \rangle_M$. This means N is an amalgam of M and B over A . For every $C \in K(A, B)$, the definition of $K(A, B)$ gives $N \models \neg \exists \bar{z} \theta_{C,A}(\bar{z}, A)$. Because $M \subset N$ and $\theta_{C,A}$ is quantifier-free, $M \models \neg \exists \bar{z} \theta_{C,A}(\bar{z}, A)$. Hence we can find $B' \subset M \models \theta_{B,A}(B', A)$ and $\bar{b}' \in B'$ such that $M \models \varphi(\bar{b}', \bar{a})$. $\square$

2.2 Graded Lie algebras and block decomposition

Definition 2.7. A graded Lie ring L is an abelian group $(L, +)$ with a decomposition $L = \bigoplus_{i \geq 1} L_i$ and a Lie bracket $[-, -] : L \times L \rightarrow L$ satisfying the following:

1. $[L_i, L_j] \subset L_{i+j}$
2. $[-, -]$ is biadditive: $[a + b, c] = [a, c] + [b, c]$ and $[a, b + c] = [a, b] + [a, c]$
3. $[a, a] = 0$
4. Jacobi identity: $[[a, b], c] + [[b, c], a] + [[c, a], b] = 0$

If L is a vector space over a field k and $[-, -]$ is k -bilinear, then L is called a graded Lie algebra over k .

Remark 2.8. 1. $[a, b] = -[b, a]$ for $a, b \in L$.

2. L is generated by L_1 if and only if $[L_i, L_1] = L_{i+1}$ for every $i \geq 1$.

Example 2.9 (Grassmann algebra). Let V be a vector space over $\mathbb{F}_3$, and set

$$\Lambda^{[1,3]}V := \Lambda^1V \oplus \Lambda^2V \oplus \Lambda^3V.$$

For homogeneous elements $u \in \Lambda^iV$ and $v \in \Lambda^jV$, define

$$[u, v] := \begin{cases} u \wedge v, & (i, j) = (1, 1) \text{ or } (2, 1), \\ -u \wedge v, & (i, j) = (1, 2), \\ 0, & i + j \geq 4. \end{cases}$$

Extending this bracket bilinearly makes $\Lambda^{[1,3]}V$ a graded Lie algebra over $\mathbb{F}_3$.

Definition 2.10. Let V be a vector space and fix a decomposition $V = \bigoplus_{\lambda} V_{\lambda}$. Each V_{λ} is called a block or λ -block. A subspace $W \subset V$ is called block-homogeneous (with respect to the decomposition) if $W = \bigoplus_{\lambda} (W \cap V_{\lambda})$.

Proposition 2.11. Let $V = \bigoplus_{\lambda} V_{\lambda}$ be a decomposition of vector space and let $W \subset V$ be a block-homogeneous subspace of V . Then $V/W \cong \bigoplus_{\lambda} V_{\lambda}/(W \cap V_{\lambda})$ with a standard isomorphism $\bar{v} \mapsto \bigoplus_{\lambda} \bar{v}_{\lambda}$ where $v = \bigoplus_{\lambda} v_{\lambda} \in V$.

Proof. Straightforward. □

2.3 Group-theoretic background

Let G be a group. For subsets $A, B \subset G$, let $[A, B]$ denote the subgroup generated by $\{[a, b] : a \in A, b \in B\}$.

Definition 2.12 (Lower and upper central series). 1. The lower central series is defined by

$$\gamma_1(G) = G \text{ and}$$

$$\gamma_{n+1}(G) = [\gamma_n(G), G] \quad (n \geq 1).$$

2. The upper central series is given by $Z_0(G) = 1$ and

$$Z_n(G) = \{a \in G : [a, g_1, \dots, g_n] = 1 \text{ for all } g_1, \dots, g_n \in G\} \quad (n \geq 1).$$

Remark 2.13. This description of Z_n is equivalent to the usual recursive definition of the upper central series. Also, this form is useful for our argument.

Fact 2.14. 1. $[\gamma_i(G), \gamma_j(G)] \leq \gamma_{i+j}(G)$, hence $\gamma_i(G)/\gamma_{i+1}(G)$ is abelian.

2. If G is nilpotent of class at most n , then $\gamma_i(G) \leq Z_{n+1-i}(G)$ for every $1 \leq i \leq n$.

In what follows, we assume groups are of exponent 3 unless otherwise noted.

Fact 2.15. Let G be an exponent 3 group and $a, b, c, d \in G$. The following are well known and elementary.

1. $[a, b, c, d] = 1$; equivalently, G is nilpotent of class at most 3.
2. $\gamma_3(G) \leq Z(G)$ and $\gamma_2(G) \leq Z_2(G)$.
3. $\gamma_2(G)$ is abelian, but $Z_2(G)$ may not be abelian in general.
4. $[a, b, c] = [b, c, a] = [c, a, b]$. In particular, $[a, b, b] = 1$; equivalently, G is 2-Engel.
5. $[a, b^{-1}] = [a, b]^{-1} = [a^{-1}, b]$
6. $[a^b, a^c] = 1$ where $a^b = b^{-1}ab$. Hence the normal closure $\langle\langle a \rangle\rangle_G$ is abelian.
7. $[a, b, c] = [b, a, c]^{-1}$
8. $[a, bc] = [a, b][a, c][a, b, c]$
9. $[ab, c] = [a, c][b, c][a, b, c]^{-1}$
10. $[A, BC] \subset [A, B][A, C][A, B, C]$
11. $[AB, C] \subset [A, C][B, C][A, B, C]$

The next fact is also obtained by standard calculations of groups.

Fact 2.16. Let $\mathcal{I}$ and $\mathcal{J}$ be disjoint sets. Then $F_{\mathcal{V}_3}(\mathcal{I} \cup \mathcal{J}) = F_{\mathcal{V}_3}(\mathcal{I}) *_{\mathcal{V}_3} F_{\mathcal{V}_3}(\mathcal{J})$.

Lemma 2.17. Let $G, H \in \mathcal{V}_3$. Then natural mappings $j_G : G \rightarrow G *_{\mathcal{V}_3} H$ and $j_H : H \rightarrow G *_{\mathcal{V}_3} H$ are injective. Hence we can regard $G, H \leq G *_{\mathcal{V}_3} H$.

Proof. Define $f : G *_{\mathcal{V}_3} H \rightarrow G$ by $f(j_H(h)) = 1$ for all $h \in H$ and $f(j_G(g)) = g$ for all $g \in G$. Then f is a retraction of j_G and hence j_G is injective. By symmetry, j_H is also injective. $\square$

We recall the associated graded Lie algebra $\text{gr}(G)$ for a group G of exponent 3.

Definition 2.18. Let G be a group of exponent 3.

1. $\text{gr}_n(G) := \gamma_n(G)/\gamma_{n+1}(G)$.

2. For $a \in \gamma_n(G)$ and $b \in \gamma_m(G)$, define

$$[a\gamma_{n+1}(G), b\gamma_{m+1}(G)] := [a, b]\gamma_{n+m+1}(G).$$

Then $\text{gr}(G) = \bigoplus_i \text{gr}_i(G)$ is a graded Lie algebra. Indeed, the inclusions $[\gamma_i(G), \gamma_j(G)] \leq \gamma_{i+j}(G)$ show that the bracket is independent of the chosen representatives. Each quotient $\text{gr}_n(G)$ has exponent 3, and hence is naturally a vector space over $\mathbb{F}_3$. The remaining axioms follow from Fact 2.15.

The following are also immediate by Fact 2.15.

Lemma 2.19. *Let G be a group of exponent 3.*

1. $\text{gr}_4(G) = 0$.
2. $[a, b, c] = [b, c, a]$ and $[a, b, b] = 0$ for every $a, b, c \in \text{gr}(G)$.
3. $\text{gr}(G)$ is generated by $\text{gr}_1(G)$ as a Lie algebra over $\mathbb{F}_3$.

Notation 2.20. 1. For $a \in \gamma_i(G)$, we define $\text{gr}_i(a) = a\gamma_{i+1}(G) \in \text{gr}_i(G)$.

2. For $K \leq G$,

$$\text{gr}_n^G(K) := \{\text{gr}_n(a) \mid a \in \gamma_n(G) \cap K\} = \frac{(K \cap \gamma_n(G))\gamma_{n+1}(G)}{\gamma_{n+1}(G)} \leq \text{gr}_n(G).$$

$$\text{Hence } \text{gr}_n^G(K) \cong (K \cap \gamma_n(G)) / (K \cap \gamma_{n+1}(G)).$$

Example 2.21. 1. If G is abelian, then $\text{gr}(G) = \text{gr}_1(G) = G$.

2. If $G = H \times K$, then $\gamma_i(G) = \gamma_i(H) \times \gamma_i(K)$ and $\text{gr}_i(G) \cong \text{gr}_i(H) \oplus \text{gr}_i(K)$.

3. Let $F_2 := F_{\mathcal{V}_3}(\{x, y\})$ be a free group in $\mathcal{V}_3$ generated by $\{x, y\}$. Then $\gamma_2(F_2) = \{[x, y]^k \mid 0 \leq k \leq 2\}$ and $\gamma_3(F_2) = 1$, hence $\text{gr}_1(F_2) = \mathbb{F}_3\bar{x} \oplus \mathbb{F}_3\bar{y}$ and $\text{gr}_2(F_2) = \mathbb{F}_3[x, y]$.

If $f : G \rightarrow H$ is a group homomorphism, then $\gamma_i(f(G)) \leq \gamma_i(H)$ holds. So we can consider an induced homomorphism between associated graded Lie algebras.

Definition 2.22. For a group homomorphism $f : G \rightarrow H$, put $\text{gr}(f) = \bigoplus_{i \geq 1} \text{gr}_i(f) : \text{gr}(G) \rightarrow \text{gr}(H)$, where

$$\text{gr}_i(f)(a\gamma_{i+1}(G)) = f(a)\gamma_{i+1}(H).$$

We can describe when $\text{gr}(f)$ is injective by using strict extensions given in the next definition.

Definition 2.23. Let $G \leq H$ be groups of exponent 3. We say that G is *strict with respect to the lower central series* in H , and write $G \leq_{\text{LCS}} H$, if

1. $\gamma_2(G) = \gamma_2(H) \cap G$ and
2. $\gamma_3(G) = \gamma_3(H) \cap G$.

Proposition 2.24. *Let $f : G \rightarrow H$ be a homomorphism of exponent 3 groups.*

1. *If $\text{gr}(f) : \text{gr}(G) \rightarrow \text{gr}(H)$ is injective then so is f .*

2. If f is injective, then the following are equivalent.

(a) $\text{gr}(f) : \text{gr}(G) \rightarrow \text{gr}(H)$ is injective.

(b) $f(G) \leq_{\text{LCS}} H$.

Proof. For (1), suppose that f is not injective and choose $1 \neq a \in \ker(f)$. Since $\gamma_4(G) = 1$, there is a largest $i \in \{1, 2, 3\}$ such that $a \in \gamma_i(G)$. Then $a\gamma_{i+1}(G)$ is a nonzero element of $\text{gr}_i(G)$ that belongs to $\ker(\text{gr}_i(f))$. Hence $\text{gr}(f)$ is not injective.

For (2), assume that f is injective and identify G with $f(G) \leq H$. For $i = 1, 2, 3$, one has

$$\ker(\text{gr}_i(f)) = \frac{\gamma_i(G) \cap \gamma_{i+1}(H)}{\gamma_{i+1}(G)}.$$

Thus $\text{gr}_1(f)$ is injective if and only if $G \cap \gamma_2(H) = \gamma_2(G)$. Under this equality, $\text{gr}_2(f)$ is injective if and only if $G \cap \gamma_3(H) = \gamma_3(G)$. Finally, $\text{gr}_3(f)$ is injective because $\gamma_4(H) = 1$. These are exactly the two conditions defining $G \leq_{\text{LCS}} H$. $\square$

Definition 2.25. Let $G \in \mathcal{V}_3$. We say that the lower central series of G coincides with the upper central series in reverse order if $\gamma_i(G) = Z_{4-i}(G)$ for every $1 \leq i \leq 3$.

Lemma 2.26. Let $G \leq H \in \mathcal{V}_3$.

1. $\gamma_i(G) \subset \gamma_i(H) \cap G \subset Z_{4-i}(H) \cap G \subset Z_{4-i}(G)$ for $1 \leq i \leq 3$.
2. If the lower central series of G coincides with the upper central series in reverse order, then $G \leq_{\text{LCS}} H$.

Proof. (1) is immediate from the definition of γ_i and Z_{4-i} , because $\gamma_4(H) = 1$.

(2) is an easy application of (1). $\square$

To determine an associated graded Lie algebra $\text{gr}(F)$ of $F = F_{\mathcal{V}_3}(\mathcal{I})$, we recall the following result:

Fact 2.27 (Levi and van der Waerden's theorem). Let $\mathcal{I} = \{x_0, \dots, x_{r-1}\}$ and $F = F_{\mathcal{V}_3}(\mathcal{I}) = B(r, 3)$ the Burnside group of r -generators and exponent 3. Then $|F| = 3^{t(r)}$ where $t(r) = r + \binom{r}{2} + \binom{r}{3}$, and every $a \in F$ is uniquely expressed as follows.

$$a = x_0^{l_0} \cdots x_{r-1}^{l_{r-1}} \prod_{i < j < r} [x_i, x_j]^{m_{i,j}} \prod_{i < j < k < r} [x_i, x_j, x_k]^{n_{i,j,k}} \quad (l_i, m_{i,j}, n_{i,j,k} \in \mathbb{F}_3)$$

Remark 2.28. When $\mathcal{I} = \{x_i \mid i < \kappa\}$ for some infinite cardinal κ , then each $a \in F_{\mathcal{V}_3}(\mathcal{I})$ has a unique expression

$$a = \prod_{i < \kappa} x_i^{l_i} \prod_{i < j < \kappa} [x_i, x_j]^{m_{i,j}} \prod_{i < j < k < \kappa} [x_i, x_j, x_k]^{n_{i,j,k}}$$

where $l_i, m_{i,j}, n_{i,j,k} \in \mathbb{F}_3$ and the coefficient families (l_i) , $(m_{i,j})$, and $(n_{i,j,k})$ have finite support. This follows from that $F_{\mathcal{V}_3}(\mathcal{I} \sqcup \mathcal{J}) = F_{\mathcal{V}_3}(\mathcal{I}) *_{\mathcal{V}_3} F_{\mathcal{V}_3}(\mathcal{J})$.

We use this normal form to identify the associated graded Lie algebra of $B(r, 3)$.

Proposition 2.29. Let F be the Burnside group $B(r, 3) = F_{\mathcal{V}_3}(\{x_0, \dots, x_{r-1}\})$. Let V be a $\mathbb{F}_3$ -vector space with basis $\{x_0, \dots, x_{r-1}\}$. Define linear maps $\sigma_n : \text{gr}_n(F) \rightarrow \Lambda^n V$ by

1. $\overline{x_i} \mapsto x_i$ for $n = 1$,
2. $\overline{[x_i, x_j]} \mapsto x_i \wedge x_j$ for $n = 2$ and $i < j$,
3. $\overline{[x_i, x_j, x_k]} \mapsto x_i \wedge x_j \wedge x_k$ for $n = 3$ and $i < j < k$,

where $\bar{a} = a\gamma_{i+1}(F) \in \text{gr}_i(F)$ for $a \in \gamma_i(F)$. Then

$$\sigma = \sigma_1 \oplus \sigma_2 \oplus \sigma_3 : \text{gr}(F) \longrightarrow \Lambda^{[1,3]}V$$

is an isomorphism of graded Lie algebras, where the bracket on $\Lambda^{[1,3]}V$ is the one defined in Example 2.9.

Proof. By Fact 2.27, the following are bases over $\mathbb{F}_3$:

$$\{\overline{x_i} : 0 \leq i < r\} \quad \text{for } \text{gr}_1(F),$$

$$\{\overline{[x_i, x_j]} : i < j\} \quad \text{for } \text{gr}_2(F),$$

and

$$\{\overline{[x_i, x_j, x_k]} : i < j < k\} \quad \text{for } \text{gr}_3(F).$$

Hence each σ_n is a linear isomorphism. The definition of the bracket gives

$$\sigma_2(\overline{[x_i, x_j]}) = x_i \wedge x_j$$

and

$$\sigma_3(\overline{[x_i, x_j, x_k]}) = x_i \wedge x_j \wedge x_k.$$

The brackets in the opposite order are determined by alternation, and all brackets of total degree at least 4 vanish. Thus σ preserves the Lie bracket. $\square$

Remark 2.30. The above result remains true when F is generated by an infinite set $\{x_i \mid i < \kappa\}$ for some cardinal number κ . See Remark 2.28.

Using this isomorphism, we identify $\text{gr}_i(B(r, 3))$ with $\Lambda^i V$ for $i = 1, 2, 3$.

Now let $G \in \mathcal{V}_3$ be a finite group generated by $\{g_i \mid 1 \leq i \leq r\} \subset G$. Then G is a quotient of $F_{\mathcal{V}_3}(\{x_1, \dots, x_r\})$, via the homomorphism induced by $x_i \mapsto g_i$. Hence, in order to understand $\text{gr}(G)$, we need to know how to calculate the associated graded Lie algebra $\text{gr}(F/N)$.

Now we calculate the associated graded Lie algebras of quotient groups.

Recall that

$$\text{gr}_i^G(N) = \frac{(N \cap \gamma_i(G))\gamma_{i+1}(G)}{\gamma_{i+1}(G)} \leq \text{gr}_i(G)$$

for $N \leq G$.

Lemma 2.31. *Let G be a group of exponent 3 and let $N \trianglelefteq G$. Then there is a canonical isomorphism $\text{gr}_i(G/N) \cong \text{gr}_i(G)/\text{gr}_i^G(N)$.*

Proof. Since $\gamma_i(G/N) = \gamma_i(G)N/N$,

$$\text{gr}_i(G/N) \cong \frac{\gamma_i(G)N}{\gamma_{i+1}(G)N} \cong \frac{\gamma_i(G)}{\gamma_i(G) \cap (\gamma_{i+1}(G)N)} = \frac{\gamma_i(G)}{\gamma_{i+1}(G)(\gamma_i(G) \cap N)} \cong \frac{\text{gr}_i(G)}{\text{gr}_i^G(N)}.$$

$\square$

3 Main results

In this section we prove that T_3 has a model companion, except a proof of Proposition A. We will give a proof of the proposition in the next section.

Proposition 3.1. *For any $a \in G \in \mathcal{V}_3$,*

$$\langle\langle a \rangle\rangle_G = \{a^{g_0} \cdots a^{g_{n-1}} \mid g_i \in G, n \leq 3\}.$$

In other words, every element in $\langle\langle a \rangle\rangle_G$ is a product of at most 3 conjugates of a .

Proof. Let $E_a = \{a^k[a, g] \mid g \in G, k \in \mathbb{F}_3\} \subset \langle\langle a \rangle\rangle$. Because $[a, g][a, g'] = [a, gg'[g, g']]$, $[a, g]^{-1} = [a, g^{-1}]$ and $a[a, g] = [a, g]a$, E_a is a subgroup. Since $a^g = g^{-1}ag = a[a^{-1}, g^{-1}] = a[a, g]$, $E_a = \langle\langle a \rangle\rangle_G$. Since $a[a, g] = a^g$, $a^2[a, g] = aa^g$, and $[a, g] = a^{-1}a^g = a^2a^g$, every element in E_a is a product of at most 3 conjugates of a . $\square$

Lemma 3.2. *Let $B, G \leq H \in \mathcal{V}_3$, $d(B) \leq m$, $\langle B, G \rangle = H$ and $\Delta \subset H$ with $|\Delta| \leq n$. Then for any $h \in \langle\langle \Delta \rangle\rangle_H$, there is $C \leq G$ such that*

1. $d(C) \leq 3(m+1)n$,
2. $h \in \langle\langle \Delta \rangle\rangle_{H_0}$ where $H_0 = \langle C, B, \Delta \rangle \leq H$.

Proof. Let $\Delta = \{\delta_i \mid i < r\}$ with $r \leq n$. Then $\langle\langle \Delta \rangle\rangle_H = \langle\langle \delta_0 \rangle\rangle_H \cdots \langle\langle \delta_{r-1} \rangle\rangle_H$, so it suffices to prove the claim for $\Delta = \{\delta\}$.

By Proposition 3.1, $h = \prod_{i < k} \delta^{u_i}$ for some $u_i \in H$ and $k \leq 3$. Since $H = \langle G, B \rangle$, every $u \in H$ is expressed as

$$u = gb \prod_{j < m} [g_j, b_j]^{k_j} \mod \gamma_3(H)$$

for some $g, g_j \in G$, $b \in B$ and $k_j \in \mathbb{F}_3$ where $B = \langle b_j \mid j < m \rangle$. Here, we used that $[gg', b] = [g, b][g', b][g, g', b]^{-1}$, $[b, g] = [g, b]^{-1}$ and $[H, H]$ is abelian. Because $\gamma_3(H) \leq Z(H)$, $\delta^u = \delta^v$ where $u = v \mod \gamma_3(H)$.

Hence C is obtained by collecting all $g, g_j \in G$ in the above expression for each u_i ($i < k$) as generators of C .

This argument shows that C is generated by at most $k(m+1)|\Delta| \leq k(m+1)n \leq 3(m+1)n$ elements. $\square$

A proof of the next proposition is given in the next section as Proposition 4.12.

Proposition A. *There is a function $f_0 : \mathbb{N} \rightarrow \mathbb{N}$ such that for every $n \in \mathbb{N}$, every e.c. model M of T_3 , and every $A \leq M$ with $d(A) \leq n$, there is $A \leq D \leq_{\text{LCS}} M$ such that D is generated by at most $f_0(n)$ elements.*

Theorem 3.3 (Main theorem). *There is a function $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfying the following: Let T_3 be the theory of the groups of exponent 3 and M an existentially closed model of T_3 . Let $A \leq M$ be a finite subgroup and let $A \leq B \in \mathcal{V}_3$ be such that $d(B) \leq m$. If M and B don't have an amalgamation in $\mathcal{V}_3$ over A , then there is $A \leq D \leq M$ such that*

1. D and B don't have an amalgamation in $\mathcal{V}_3$ over A
2. $d(D) \leq f(m)$ element.

Proof. Suppose that $A \leq M$ and $A \leq B$ don't have an amalgam over A . Let $H = M *_{\mathcal{V}_3} B$ and let $j_M : M \rightarrow H$ and $j_B : B \rightarrow H$ be canonical embeddings. Let $d(A) = n$ and let A be generated by $\{a_0, \dots, a_{n-1}\}$. Put $\Delta = \{\delta_{a_i} \mid i < n\} \subset H$ where $\delta_a = j_M(a)j_B(a)^{-1}$. By the assumption, at least one of the canonical homomorphisms $\bar{j}_M : M \rightarrow H/\langle\langle \Delta \rangle\rangle_H$ and $\bar{j}_B : B \rightarrow H/\langle\langle \Delta \rangle\rangle_H$ is not injective; equivalently, at least one of the following holds:

- There is $1 \neq g \in M$ such that $j_M(g) \in \langle\langle \Delta \rangle\rangle_H$.
- There is $1 \neq g \in B$ such that $j_B(g) \in \langle\langle \Delta \rangle\rangle_H$.

Since the other case is similar, we assume $g \in M$ and $j_M(g) \in \langle\langle \Delta \rangle\rangle_H$.

Recall that $d(A) = n$ and $d(B) \leq m$. By Lemma 3.2, there is $C_0 \leq M$ generated by $3(m+1)n$ elements such that, by letting $H_0 = \langle j_M(C_0), j_B(B), \Delta \rangle \leq H$, $j_M(g) \in \langle\langle \Delta \rangle\rangle_{H_0}$. Let $C = \langle C_0, A, g \rangle_M$, which is generated by $3(m+1)n + n + 1$ elements. Since $n = d(A) \leq \log_3(|A|) \leq \log_3(|B|) \leq t(m)$ where $t(m) = m + \binom{m}{2} + \binom{m}{3}$, we have $3(m+1)n + n + 1 \leq (3m+4)t(m) + 1$. By Proposition A, we can find $C \leq D \leq_{\text{LCS}} M$ such that D is generated by at most $f(m) := f_0((3m+4)t(m) + 1)$ elements.

Claim A. D and B don't have an amalgam in $\mathcal{V}_3$ over A .

Since $D \leq_{\text{LCS}} M$, a natural homomorphism $D *_{\mathcal{V}_3} B \rightarrow M *_{\mathcal{V}_3} B$ is an embedding, so we identify $D *_{\mathcal{V}_3} B$ as a subgroup of $M *_{\mathcal{V}_3} B$. Since $j_M|_D$ coincides a natural embedding $j_D : D \rightarrow D *_{\mathcal{V}_3} B \leq M *_{\mathcal{V}_3} B$ and $H_0 \leq \langle j_D(D), j_B(B) \rangle \leq D *_{\mathcal{V}_3} B$, we know $1 \neq j_D(g) = j_M(g) \in \langle\langle \Delta \rangle\rangle_{D *_{\mathcal{V}_3} B} \leq D *_{\mathcal{V}_3} B$. Hence $D \rightarrow D *_{\mathcal{V}_3} B / \langle\langle \Delta \rangle\rangle_{D *_{\mathcal{V}_3} B}$ is not an embedding. This shows that $A \leq D$ and $A \leq B$ don't have an amalgam over A . $\square$

By Fact 2.6, we have the following.

Corollary 3.4. *Let T_3 be the theory of the groups of exponent 3. Then T_3 has a model companion.*

4 Structural analysis

We first construct a relatively free presentation of G .

Let G be a group of exponent 3. Since $V_G = G/\gamma_2(G)$ is abelian, it is a vector space over $\mathbb{F}_3$. Let $\mathcal{I} = \{e_i \mid i \in I\}$ be a basis of V_G . For each $i \in I$, choose a representative $a_i \in G$ such that $e_i = a_i\gamma_2(G)$.

Proposition 4.1. *The assignment $e_i \mapsto a_i$ induces a surjective homomorphism $\varphi_G : F_{\mathcal{V}_3}(\mathcal{I}) \rightarrow G$ such that $\ker(\varphi_G) \leq \gamma_2(F_{\mathcal{V}_3}(\mathcal{I}))$.*

Proof. Let $F = F_{\mathcal{V}_3}(\mathcal{I})$ and $H = \langle a_i \mid i \in I \rangle \leq G$. Since the cosets $a_i\gamma_2(G)$ form a basis of $G/\gamma_2(G)$, we have $G = H\gamma_2(G)$.

We claim that $H = G$. Since $\gamma_2(G)$ is abelian and $\gamma_3(G) \leq Z(G)$, the commutator identities give

$$\gamma_2(G) = [H\gamma_2(G), H\gamma_2(G)] \leq H[H, \gamma_2(G)] \leq H\gamma_3(G).$$

Consequently,

$$[H, \gamma_2(G)] \leq [H, H\gamma_3(G)] = [H, H] \leq H,$$

where we used the centrality of $\gamma_3(G)$. It follows that $\gamma_2(G) \leq H[H, \gamma_2(G)] \leq H$, and hence $G = H\gamma_2(G) = H$. Thus φ_G is surjective.

The induced map

$$\bar{\varphi}_G : F/\gamma_2(F) \longrightarrow G/\gamma_2(G)$$

sends the basis $\mathcal{I}$ bijectively to the basis $\{a_i\gamma_2(G) : i \in I\}$, so it is an isomorphism. If $a \in \ker(\varphi_G)$, then $a\gamma_2(F) \in \ker(\bar{\varphi}_G) = 0$. Therefore $a \in \gamma_2(F)$, as required. $\square$

Lemma 4.2. *Let G be a group of exponent 3, let $K \leq \gamma_2(G)$, and let $L = \langle\langle K \rangle\rangle_G$ be the normal closure of K in G . Then*

1. $L = K[K, G]$ and $L \leq \gamma_2(G)$,
2. $\text{gr}_2^G(L) = \text{gr}_2^G(K) = \{\text{gr}_2(a) \in \text{gr}_2(G) \mid a \in K \cap \gamma_2(G)\}$,
3. $\text{gr}_3^G(L) = \text{gr}_3^G(K) + [\text{gr}_2^G(K), \text{gr}_1(G)]$.

Proof. Since $K \leq \gamma_2(G)$, we have $[K, G] \leq \gamma_3(G) \leq Z(G)$. For $k \in K$ and $g \in G$,

$$k^g = [g^{-1}, k]k \in K[K, G].$$

Thus every conjugate of an element of K belongs to $K[K, G]$. The reverse inclusion follows because the normal closure of K contains both K and $[K, G]$. Hence

$$L = K[K, G] \leq \gamma_2(G).$$

Since $[K, G] \leq \gamma_3(G)$, the images of L and K in $\text{gr}_2(G) = \gamma_2(G)/\gamma_3(G)$ coincide. This proves (2).

Moreover,

$$L \cap \gamma_3(G) = (K \cap \gamma_3(G))[K, G].$$

For $k \in K$ and $g \in G$, the induced bracket is

$$[k\gamma_3(G), g\gamma_2(G)] = [k, g]\gamma_4(G) = [k, g].$$

Therefore the image of $[K, G]$ in $\text{gr}_3(G)$ is exactly $[\text{gr}_2^G(K), \text{gr}_1(G)]$, which proves (3). $\square$

The next proposition is a main tool to analyze group structures of exponent 3 in this section.

Proposition 4.3. *Let G_i ($i = 0, 1$) be groups of exponent 3.*

1. *Then the following natural mappings induced from the canonical embeddings $G_i \rightarrow G_0 *_{\nu_3} G_1$ ($i = 0, 1$) are isomorphisms.*

$$\eta_1 : \text{gr}_1(G_0) \oplus \text{gr}_1(G_1) \rightarrow \text{gr}_1(G_0 *_{\nu_3} G_1)$$

$$\eta_2 : \text{gr}_2(G_0) \oplus (\text{gr}_1(G_0) \otimes \text{gr}_1(G_1)) \oplus \text{gr}_2(G_1) \rightarrow \text{gr}_2(G_0 *_{\nu_3} G_1)$$

$$\eta_3 : \text{gr}_3(G_0) \oplus (\text{gr}_2(G_0) \otimes \text{gr}_1(G_1)) \oplus (\text{gr}_1(G_0) \otimes \text{gr}_2(G_1)) \oplus \text{gr}_3(G_1) \rightarrow \text{gr}_3(G_0 *_{\nu_3} G_1)$$

$$\text{where } \text{gr}_i(G_0) \otimes \text{gr}_j(G_1) \ni a \otimes b \xrightarrow{\eta_{i+j}} [\eta_i(a), \eta_j(b)] \in \text{gr}_{i+j}(G_0 *_{\nu_3} G_1).$$

2. Let us identify $\text{dom}(\eta_i)$ with $\text{Im}(\eta_i)$. Then, by letting $\mathcal{B}_{i,j} = \text{gr}_i(G_0) \otimes \text{gr}_j(G_1)$ for $i, j \geq 1$ with $i + j \leq 3$, $\mathcal{B}_{i,0} = \text{gr}_i(G_0)$, $\mathcal{B}_{0,j} = \text{gr}_j(G_1)$, and $\mathcal{B}_{i,j} = 0$ ($i + j \geq 4$),

$$[\mathcal{B}_{i,j}, \mathcal{B}_{k,l}] \subset \mathcal{B}_{i+k,j+l}.$$

Proof. (2) is immediate from the definition of η_i . We show (1). Let $V_n = \text{gr}_1(G_n)$ ($n < 2$) and fix a basis $\mathcal{I}_n$ of V_n and a representative $a_e \in G_n$ of each vector $e \in \mathcal{I}_n$. Let us consider a lift $F_{G_n} := F_{\mathcal{V}_3}(\mathcal{I}_n)$ of G_n , and we will use the chosen surjective homomorphism $\varphi_n : F_{G_n} \rightarrow G_n$ and its kernel $K_n := \ker(\varphi_n) \leq \gamma_2(F_{G_n})$ as defined in Proposition 4.1. Let $V = V_0 \oplus V_1$, $F = F_{G_0} *_{\mathcal{V}_3} F_{G_1} = F_{\mathcal{V}_3}(\mathcal{I}_0 \sqcup \mathcal{I}_1)$, $K = \langle K_0, K_1 \rangle_F = K_0 K_1$, and $L = \langle\langle K \rangle\rangle_F$. (Notice that $\text{gr}_1(F) = V$.) Then we can identify $G_0 *_{\mathcal{V}_3} G_1$ with F/L .

By Lemma 2.31, we have a canonical isomorphism $\text{gr}_i(F/L) \cong \text{gr}_i(F)/\text{gr}_i^F(L)$. Also by Lemma 4.2,

$$\text{gr}_1^F(L) = 0, \quad \text{gr}_2^F(L) = \text{gr}_2^F(K), \quad \text{gr}_3^F(L) = \text{gr}_3^F(K) + [\text{gr}_2^F(K), \text{gr}_1(F)].$$

Now recall that we can identify $\text{gr}_i(F)$ with $\Lambda^i V$ and recall the standard direct sum decomposition of Grassmann algebra:

$$\begin{aligned} \Lambda^1(V_0 \oplus V_1) &= \Lambda^1 V_0 \oplus \Lambda^1 V_1 = V_0 \oplus V_1, \\ \Lambda^2(V_0 \oplus V_1) &= \Lambda^2 V_0 \oplus (V_0 \otimes V_1) \oplus \Lambda^2 V_1, \\ \Lambda^3(V_0 \oplus V_1) &= \Lambda^3 V_0 \oplus (\Lambda^2 V_0 \otimes V_1) \oplus (V_0 \otimes \Lambda^2 V_1) \oplus \Lambda^3 V_1. \end{aligned}$$

Hence,

$$\begin{aligned} \text{gr}_1(G_0 *_{\mathcal{V}_3} G_1) &\cong \text{gr}_1(F)/\text{gr}_1^F(L) \\ &= V_0 \oplus V_1. \end{aligned}$$

Since $K = K_0 K_1$ is abelian, by letting $\text{gr}_2^F(K_n) = R_n \subset \Lambda^2 V_n$,

$$\text{gr}_2^F(K) = R_0 \oplus R_1 \subset \Lambda^2 V_0 \oplus \Lambda^2 V_1 \subset \Lambda^2 V.$$

Hence $\text{gr}_2^F(K)$ is block-homogeneous with respect to the decomposition. After we identify $\text{gr}_2(F_{G_n})$ with $\Lambda^2 V_n$, we can consider $\text{gr}_2^{F_{G_n}}(K_n) = \text{gr}_2^F(K_n) = R_n$. Hence, because $\Lambda^2 V_n/R_n \cong \text{gr}_2(F_{G_n}/K_n) = \text{gr}_2(G_n)$,

$$\begin{aligned} \text{gr}_2(G_0 *_{\mathcal{V}_3} G_1) &\cong \text{gr}_2(F)/\text{gr}_2^F(L) \\ &= \Lambda^2(V_0 \oplus V_1)/\text{gr}_2^F(K) \\ &\cong \frac{\Lambda^2 V_0 \oplus (V_0 \otimes V_1) \oplus \Lambda^2 V_1}{R_0 \oplus R_1} \\ &\cong \text{gr}_2(G_0) \oplus (\text{gr}_1(G_0) \otimes \text{gr}_1(G_1)) \oplus \text{gr}_2(G_1). \end{aligned}$$

Let $S = \text{gr}_3^F(L) \subset \Lambda^3 V$, $S_n = \text{gr}_3^{F_{G_n}}(K_n) \subset \Lambda^3 V_n$. Then $\text{gr}_3^F(K) = S_0 \oplus S_1$. By similar arguments as above,

$$S = S_0 + S_1 + [R_0 + R_1, V] = S_0 + S_1 + (R_0 + R_1) \wedge (V_0 + V_1).$$

Because $R_n \wedge V_n \subset S_n$, we have

$$S = S_0 + R_0 \wedge V_1 + V_0 \wedge R_1 + S_1, \quad R_n \wedge V_{1-n} \cong R_n \otimes V_{1-n}.$$

Hence,

$$\begin{aligned}
\text{gr}_3(G_0 *_{\mathcal{V}_3} G_1) &\cong \text{gr}_3(F) / \text{gr}_3^F(L) \\
&\cong \frac{\Lambda^3(V_0) \oplus (\Lambda^2 V_0 \otimes V_1) \oplus (V_0 \otimes \Lambda^2 V_1) \oplus \Lambda^3 V_1}{S_0 + R_0 \otimes V_1 + V_0 \otimes R_1 + S_1} \\
&\cong \text{gr}_3(G_0) \oplus (\text{gr}_2(G_0) \otimes \text{gr}_1(G_1)) \oplus (\text{gr}_1(G_0) \otimes \text{gr}_2(G_1)) \oplus \text{gr}_3(G_1),
\end{aligned}$$

because $S = S_0 + R_0 \otimes V_1 + V_0 \otimes R_1 + S_1$ is block-homogeneous. $\square$

In the remainder of this section, we will often consider $H = G *_{\mathcal{V}_3} F_n$ where $F_n = F_{\mathcal{V}_3}(x_1, \dots, x_n)$. By Proposition 4.3, we use the following identifications:

$$\begin{aligned}
\text{gr}_1(H) &= \text{gr}_1(G) \oplus \text{gr}_1(F_n) \\
\text{gr}_2(H) &= \text{gr}_2(G) \oplus (\text{gr}_1(G) \otimes \text{gr}_1(F_n)) \oplus \text{gr}_2(F_n) \\
\text{gr}_3(H) &= \text{gr}_3(G) \oplus (\text{gr}_2(G) \otimes \text{gr}_1(F_n)) \oplus (\text{gr}_1(G) \otimes \text{gr}_2(F_n)) \oplus \text{gr}_3(F_n)
\end{aligned}$$

and, by Proposition 2.29,

$$\text{gr}_i F_n = \Lambda^i V_n$$

where V_n is a vector space over $\mathbb{F}_3$ with basis $\{x_j \mid 1 \leq j \leq n\}$.

Lemma 4.4. *If $D \leq_{\text{LCS}} G \in \mathcal{V}_3$, then for every $B \in \mathcal{V}_3$, a natural homomorphism $D *_{\mathcal{V}_3} B \rightarrow G *_{\mathcal{V}_3} B$ induced by the inclusion $D \leq G$ is injective.*

Proof. Since $D \leq_{\text{LCS}} G$, a natural homomorphism $\text{gr}(D) \rightarrow \text{gr}(G)$ is injective by Proposition 2.24. Hence, by Proposition 4.3, $\text{gr}(D *_{\mathcal{V}_3} B) \rightarrow \text{gr}(G *_{\mathcal{V}_3} B)$ is injective. Again, by Proposition 2.24, a natural homomorphism $D *_{\mathcal{V}_3} B \rightarrow G *_{\mathcal{V}_3} B$ is injective. $\square$

Lemma 4.5. *Let $1 \neq G \in \mathcal{V}_3$ and $F_2 = F_{\mathcal{V}_3}(x, y)$ be a free group generated by 2 elements. Then $G \leq_{\text{LCS}} G *_{\mathcal{V}_3} F_2$, and the lower central series of $G *_{\mathcal{V}_3} F_2$ coincides with its upper central series in reverse order.*

Proof. Let $H = G *_{\mathcal{V}_3} F_2$. Let $f : G \rightarrow H$ be a natural inclusion map. By Proposition 4.3, $\text{gr}(f) : \text{gr} G \rightarrow \text{gr} H$ is injective. Hence $G \leq_{\text{LCS}} H$ by Proposition 2.24.

Claim A. $\gamma_2(H) = Z_2(H)$.

Let $a \in H \setminus \gamma_2(H)$. We show that there is $b, c \in H$ such that $[a, b, c] \neq 1$, which implies $a \notin Z_2(H)$. Because $\text{gr}_1(H) = \text{gr}_1(G) \oplus \text{gr}_1(F_2)$, $\text{gr}_1(a) = \alpha + \beta$ for some $\alpha \in \text{gr}_1(G)$ and $\beta \in \text{gr}_1(F_2) = \Lambda^1 V_2$ where V_2 is a $\mathbb{F}_3$ -vector space with a basis $\{x, y\}$.

Case 1: Suppose $\alpha \neq 0$. Then $\text{gr}_3([a, x, y]) = -\alpha \otimes (x \wedge y) + \beta \wedge x \wedge y \neq 0$, hence $[a, x, y] \neq 1$.

Case 2: Suppose $\alpha = 0$ and $\beta \neq 0$. Take $g \in G \setminus \gamma_2(G)$. Since $\dim \text{gr}_1(F_2) = 2$, we can find $w \in F_2$ such that $\{\beta, \text{gr}_1(w)\}$ is linearly independent in $\text{gr}_1(F_2)$. Then $[a, g, w] \neq 1$ because $\text{gr}_3([a, g, w]) = \text{gr}_1(g) \otimes (\beta \wedge \text{gr}_1(w)) \neq 0$.

Claim B. $\gamma_3(H) = Z(H)$

Let $a \notin \gamma_3(H)$. We show that there is $b \in H$ such that $[a, b] \neq 1$, which implies $a \notin Z(H)$. Because of Claim A, we can assume $a \in \gamma_2(H)$. Since $\text{gr}_2(H) = \text{gr}_2(G) \oplus (\text{gr}_1(G) \otimes \text{gr}_1(F_2)) \oplus \text{gr}_2(F_2)$, we have $\text{gr}_2(a) = \alpha_2 + \alpha_1 \otimes x + \alpha'_1 \otimes y + \beta_2$ for some $\alpha_2 \in \text{gr}_2(G)$, $\alpha_1, \alpha'_1 \in \text{gr}_1(G)$, and $\beta_2 \in \text{gr}_2(F_2)$. (Here, we used that $\text{gr}_1(F_2) = \mathbb{F}_3 x \oplus \mathbb{F}_3 y$.)

Case 1: If $\alpha_2 \neq 0$, then $[a, x] \neq 1$.

Case 2: If $\alpha_2 = 0$ and $\alpha_1 \neq 0$, then $\text{gr}_3([a, y]) = -\alpha_1 \otimes (x \wedge y) + \beta_2 \wedge y = -\alpha_1 \otimes (x \wedge y) \neq 0$. In the case of $\alpha'_1 \neq 0$, use x insted of y .

Case 3: If $\alpha_2 = 0$, $\alpha_1 = \alpha'_1 = 0$ and $\beta_2 \neq 0$, then choose $g \in G \setminus \gamma_2(G)$. We have $[a, g] \neq 1$ because $\text{gr}_3([a, g]) = -\text{gr}_1(g) \otimes \beta_2 \neq 0$. $\square$

Lemma 4.6. *Let $G \in \mathcal{V}_3$, $g_1, \dots, g_n \in \gamma_2(G)$ and $F_{2n} = F_{\mathcal{V}_3}(x_1, y_1, \dots, x_n, y_n)$. Then a natural homomorphism $G \rightarrow H$ is injective where*

$$H = (G *_{\mathcal{V}_3} F_{2n}) / \langle\langle g_i^{-1}[x_i, y_i] \mid 1 \leq i \leq n \rangle\rangle.$$

Proof. Let

$$H_0 = G *_{\mathcal{V}_3} F_{2n}, \quad r_i = g_i^{-1}[x_i, y_i] \in H_0$$

$$K = \langle r_i \mid i \leq n \rangle_{H_0} \leq \gamma_2(H_0), \quad L = \langle\langle K \rangle\rangle_{H_0} = \langle\langle r_1 \rangle\rangle_{H_0} \cdots \langle\langle r_n \rangle\rangle_{H_0}, \quad H = H_0/L.$$

Claim A. $L \leq \gamma_2(H_0)$ and $\langle\langle r_i \rangle\rangle = \{r_i^m[r_i, h] \mid m \in \mathbb{F}_3, h \in H_0\}$.

Since $L = K[K, H_0]$ and $K \leq \gamma_2(H_0)$, we have $L \leq \gamma_2(H_0)$. Similarly,

$$\langle\langle r_i \rangle\rangle = \langle r_i \rangle[\langle r_i \rangle, H_0] = \langle r_i \rangle[r_i, H_0].$$

Here, we used that $[r^{-1}, h] = [r, h]^{-1} = [r, h^{-1}]$. Since $r_i \in \gamma_2(H_0)$, $[r_i, h][r_i, h'] = [r_i, hh']$. Hence $[r_i, H_0] = \{[r_i, h] \mid h \in H_0\}$.

It suffices to show that $L \cap G = 1$, so let $a \in L \cap G$. By Claim A, $a = \prod_i r_i^{m_i}[r_i, h_i] \in \gamma_2(H_0)$ for some $h_i \in H_0$.

Claim B. $m_i = 0$ ($i \leq n$). Hence $a = \prod_i [r_i, h_i] \in \gamma_3(H_0)$.

Since $g_i, r_i, a \in \gamma_2(H_0)$, we can calculate that

$$\begin{aligned} \text{gr}_2(a) &= \sum_i m_i \text{gr}_2(r_i) \\ &= \sum_i m_i (\text{gr}_2([x_i, y_i]) - \text{gr}_2(g_i)) \\ &= \sum_i m_i (x_i \wedge y_i) - \sum_i m_i \text{gr}_2(g_i). \end{aligned}$$

Because $a \in G$, $\text{gr}_2(a)$ must be in $\text{gr}_2(G)$ whereas $0 \neq x_i \wedge y_i \in \text{gr}_2(F_{2n})$ and $\{x_i \wedge y_i \mid i \leq n\}$ is linearly independent. Hence we have $m_i = 0$ for all i and it follows that $a \in \gamma_3(H_0)$.

Claim C. $a = 1$.

Recall that $a = \prod_i [r_i, h_i] \in \gamma_3(H_0)$. Suppose $\text{gr}_1(h_i) = s_i + t_i$ where $s_i \in \text{gr}_1(G)$ and $t_i \in \text{gr}_1(F_{2n})$. Then,

$$\begin{aligned} \text{gr}_3(a) &= \sum_i [x_i \wedge y_i - \text{gr}_2(g_i), s_i + t_i] \\ &= \sum_i -s_i \otimes (x_i \wedge y_i) + x_i \wedge y_i \wedge t_i - [\text{gr}_2(g_i), s_i] - \text{gr}_2(g_i) \otimes t_i. \end{aligned}$$

Because $a \in G$ and $\text{gr}_3(a) \in \text{gr}_3(G)$, $\text{gr}_3(a)$ must be $-\sum_i [\text{gr}_2(g_i), s_i]$. Hence $\sum_i s_i \otimes (x_i \wedge y_i) = 0$, which implies $s_i = 0$. Therefore we have $\text{gr}_3(a) = 0$ and $a = 1$. $\square$

Lemma 4.7. *Let $C \leq G \in \mathcal{V}_3$ such that C is generated by m elements. Then there is an extension $G \leq H \in \mathcal{V}_3$ and $C \leq D \leq H$ such that*

1. D is generated by C and at most $2m$ elements in H , and
2. $D \cap \gamma_2(H) = \gamma_2(D)$.

Proof. Let $g_1, \dots, g_n \in C \cap \gamma_2(G)$ be such that $\{g_i \gamma_2(C) \mid i \leq n\}$ is a basis of $(C \cap \gamma_2(G))/\gamma_2(C)$. Then $C \cap \gamma_2(G) \subset \langle g_i \mid i \leq n \rangle \gamma_2(C)$ and $n \leq \dim \text{gr}_1(C) \leq m$. Let $F_{2n} = F_{\mathcal{V}_3}(x_1, y_1, \dots, x_n, y_n)$, $H_0 = G *_{\mathcal{V}_3} F_{2n}$ and $H = H_0/L$ where $L = \langle\langle g_i^{-1}[x_i, y_i] \mid i \leq n \rangle\rangle$. Then, by Lemma 4.6, $G \leq H$.

Claim A. $D = \langle C, x_i, y_i \mid i \leq n \rangle_H \leq H$ satisfies that $D \cap \gamma_2(H) = \gamma_2(D)$.

First notice that, since $L \leq \gamma_2(H_0)$, $\text{gr}_1(H) = \text{gr}_1(H_0/L) \cong \text{gr}_1(H_0) = \text{gr}_1(G) \oplus \text{gr}_1(F_{2n})$.

Let $a \in D \cap \gamma_2(H)$. Then $a \in c \prod x_i^{m_i} y_i^{n_i} \gamma_2(D)$ for some $c \in C$. Hence, $0 = \text{gr}_1(a) = \text{gr}_1(c) + \sum_i m_i x_i + \sum_i n_i y_i$. Because $\{x_i, y_i \mid i \leq n\}$ is a basis of $\text{gr}_1(F_{2n})$, $m_i = n_i = 0$ for all $i \leq n$ and $\text{gr}_1(c) = 0$. Therefore $c \in \gamma_2(G)$ follows. This argument shows that $D \cap \gamma_2(H) = (C \cap \gamma_2(G)) \gamma_2(D)$. By the choice of g_i ($i \leq n$),

$$\begin{aligned} D \cap \gamma_2(H) &= (C \cap \gamma_2(G)) \gamma_2(D) \\ &\subset (\langle g_i \mid i \leq n \rangle \gamma_2(C)) \gamma_2(D) \\ &\subset \gamma_2(D). \end{aligned}$$

□

Lemma 4.8. *Let $G \in \mathcal{V}_3$, $g_1, \dots, g_n \in Z(G)$ and let $F_{3n} = F_{\mathcal{V}_3}(x_1, y_1, z_1, \dots, x_n, y_n, z_n)$. Then the natural map $G \rightarrow H$ is injective where*

$$H = (G \times F_{3n}) / \langle\langle g_i^{-1}[x_i, y_i, z_i] \mid i \leq n \rangle\rangle$$

Proof. Let $L = \langle\langle g_i^{-1}[x_i, y_i, z_i] \mid i \leq n \rangle\rangle \leq G \times F_{3n}$. Since $g_i \in Z(G)$, $g_i \in Z(G \times F_{3n})$. Hence $L = \langle g_i^{-1}[x_i, y_i, z_i] \mid i \leq n \rangle$.

We prove that $G \cap L = 1$ in $G \times F_{3n}$. Let $a \in G \cap L$. Then $a = \prod_i g_i^{-m_i} [x_i, y_i, z_i]^{m_i}$. Since $a \in G$, $m_i = 0$ for all $i \leq n$. This means $a = 1$. □

Lemma 4.9. *Let $C \leq G \in \mathcal{V}_3$ and let C be generated by at most n elements. Suppose that $C \cap \gamma_2(G) = \gamma_2(C)$. Then there are $G \leq H$ and $C \leq D \leq H$ such that*

1. D is generated by C and at most $3n(n-1)/2$ elements in H ,
2. $D \leq_{\text{LCS}} H$.

Proof. Let $g_0, \dots, g_{m-1} \in C \cap \gamma_3(G)$ be such that $\{g_i \gamma_3(C) \mid i < m\}$ generates the $\mathbb{F}_3$ -vector space $(C \cap \gamma_3(G))/\gamma_3(C)$. Since $C \cap \gamma_3(G) \leq C \cap \gamma_2(G) = \gamma_2(C)$, we can choose $m \leq \dim \text{gr}_2(C) \leq \binom{n}{2}$. By Lemma 4.8, $H = (G \times F_{3m})/L$ is an extension of G where $L = \langle g_i^{-1}[x_i, y_i, z_i] \mid i < m \rangle$. Let $D = \langle C, x_i, y_i, z_i \mid i < m \rangle_H$. Then $D = (C \times F_{3m})/L$.

Claim A. $\gamma_2(H) \cap D = \gamma_2(D)$

Since $g_i \in C \cap \gamma_3(G) \leq \gamma_2(C)$, $L \leq \gamma_2(C \times F_{3m})$. Also, $\gamma_2(H) = (\gamma_2(G \times F_{3m})L)/L = \gamma_2(G \times F_{3m})/L = (\gamma_2(G) \times \gamma_2(F_{3m}))/L$. Hence

$$\begin{aligned} \gamma_2(H) \cap D &= \frac{\gamma_2(G) \times \gamma_2(F_{3m})}{L} \cap \frac{C \times F_{3m}}{L} \\ &= \frac{(\gamma_2(G) \cap C) \times \gamma_2(F_{3m})}{L} \\ &= \frac{\gamma_2(C) \times \gamma_2(F_{3m})}{L} \\ &= \frac{\gamma_2(C \times F_{3m})}{L} \\ &= \gamma_2(D). \end{aligned}$$

Claim B. $\gamma_3(H) \cap D = \gamma_3(D)$

Since $g_i \in \gamma_3(G) \cap C$ and $[x_i, y_i, z_i] \in \gamma_3(F_{3m})$, $L \leq (\gamma_3(G) \cap C) \times \gamma_3(F_{3m}) \leq \gamma_3(G \times F_{3m})$. Hence,

$$\begin{aligned} \gamma_3(H) &= \frac{\gamma_3(G \times F_{3m})}{L}, \\ \gamma_3(H) \cap D &= \frac{(\gamma_3(G) \cap C) \times \gamma_3(F_{3m})}{L}, \\ \gamma_3(D) &= \frac{(\gamma_3(C) \times \gamma_3(F_{3m}))L}{L}. \end{aligned}$$

Suppose $a \in (\gamma_3(G) \cap C) \times \gamma_3(F_{3m})$. By the choice of g_i ($i < m$), $a = h \prod_i g_i^{m_i} f$ for some $h \in \gamma_3(C)$, $m_i \in \mathbb{F}_3$, and $f \in \gamma_3(F_{3m})$. Therefore, $a \in h(\prod_i [x_i, y_i, z_i]^{m_i})fL$. This shows that $\gamma_3(H) \cap D \subset \gamma_3(D)$. $\square$

Remark 4.10. The bound on the number of additional generators in the above lemma is not optimal. Indeed, for example, $F_{\mathcal{V}_3}(x_1, \dots, x_4)$ contains $\binom{4}{3} = 4$ elements $[x_i, x_j, x_k]$ ($i < j < k \leq 4$) that are linearly independent in $\text{gr}_3(F_4)$. Hence, for $g_1, \dots, g_4$, $L = \langle g_1^{-1}[x_1, x_2, x_3], \dots, g_4^{-1}[x_2, x_3, x_4] \rangle$ is enough to prove the desired conditions. This trick works for Lemma 4.7.

Proposition 4.11. *Let M be an e.c. model of T_3 . Then the lower central series of M coincides with its upper central series in reverse order. Moreover,*

$$\gamma_2(M) = \{[a, b] \mid a, b \in M\},$$

$$\gamma_3(M) = \{[a, b, c] \mid a, b, c \in M\}.$$

Proof. Let $a \in \gamma_2(M)$. Then there are $b', c' \in H \geq M$ such that $[b', c'] = a$ in H . Since M is e.c., we can find $b, c \in M$ such that $[b, c] = a$ in M . This proves $\gamma_2(M) = \{[a, b] \mid a, b \in M\}$. Similar argument works for $\gamma_3(M)$.

Let $a \in M \setminus \gamma_2(M)$ and let $H = M *_{\mathcal{V}_3} F_2$. Since $M \leq_{\text{LCS}} H$, $a \notin \gamma_2(H)$. Hence there are $b', c' \in H$ such that $[a, b', c'] \neq 1$. Since M is an e.c. model, we can find $b, c \in M$ such that $[a, b, c] \neq 1$, which proves $\gamma_2(M) = Z_2(M)$. Similar argument works for proving $\gamma_3(M) = Z(M)$. $\square$

Proposition 4.12. *Let M be an e.c. model of T_3 and let $C \leq M$ be a finite subgroup. Then there is $C \leq D \leq M$ such that*

1. *D is generated by at most $15n^2$ elements where C is generated by at most n elements,*
2. *the lower central series of D coincides with its upper central series in reverse order. In particular, $D \leq_{\text{LCS}} M$.*

Proof. We assume $C \neq 1$ since otherwise are trivial. By Lemma 4.7, there is an extension $M \leq H_1$, $C \leq D_1 \leq H_1$ such that $D_1 \cap \gamma_2(H_1) = \gamma_2(D_1)$. The group D_1 is generated by C and at most $2n$ elements. Hence D_1 is generated by at most $3n$ elements. Next we apply Lemma 4.9 and get an extension $H_1 \leq H_2$ and $D_1 \leq D_2 \leq_{\text{LCS}} H_2$. D_2 is generated by D_1 and at most $3(3n)(3n-1)/2$ elements. Hence D_2 is generated by at most $3n + 9n(3n-1)/2$ elements.

Let $F_2 = F_{\mathcal{V}_3}(\{x, y\})$ and set

$$D_3 = D_2 *_{\mathcal{V}_3} F_2, \quad H = H_2 *_{\mathcal{V}_3} F_2.$$

Since $1 \neq D_2 \leq_{\text{LCS}} H_2$, Lemma 4.4 implies that $D_3 \leq H$. By Lemma 4.5, the lower central series of D_3 coincides with its upper central series in reverse order. D_3 is generated by at most

$$3n + \frac{9n(3n-1)}{2} + 2 \leq 15n^2$$

elements. Since M is an e.c. model and D_3 is finite, we can find $C \leq D \leq M$ such that $D \cong_C D_3$. Since D is a copy of D_3 , the lower central series of D coincides with its upper central series in reverse order. $\square$

5 Applications and examples

Explain consequences of the main theorem, discuss examples, and indicate the scope of the hypotheses.

6 Further questions

Record natural open problems and possible extensions.

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